Sérgio Ricardo Tornich Watanabe

PHYSICS • MACHINE LEARNING • SCIENTIFIC COMPUTING • AI EVALUATION

Portland, OR • +1 (503) 754 7566 • sergiotornichwatanabe@gmail.com • github.com/SergioWatanabe • LinkedIn

Physics student at Reed College focused on machine learning, scientific computing and AI evaluation. Hands-on experience designing scientific benchmark tasks for frontier coding agents and building physics-constrained neural models in Python and PyTorch. Seeking a Summer 2027 internship where quantitative reasoning, modeling and software engineering intersect.

WORK AUTHORIZATION F-1 student, eligible for summer internships through Curricular Practical Training (CPT). No visa sponsorship required.

EDUCATION

Reed College, Portland, OR • B.A. in Physics • Expected May 2030

Raffles American School • High School Diploma, May 2025 • GPA 4.5/5 weighted • SAT 1480 • AP Physics, Calculus BC, Computer Science

EXPERIENCE

DataAnnotation | Remote • Mar 2026 to Sep 2026

AI Model Evaluation and Task Design, Physics and Scientific Computing

- Designed scientific computing regression tasks to evaluate frontier LLM coding agents using simulation and real-world physical datasets sized for laptop-scale runs.
- Set task targets through model ladders spanning linear and polynomial baselines, gradient-boosted trees and PyTorch MLPs, adding physical constraints such as mass conservation.
- Wrote oracle solutions and NumPy/PyTorch verifiers, then ran adversarial cheat trials to detect fabricated outputs and verifier manipulation.
- Submitted tasks on which frontier coding agents averaged a 0 to 16% pass rate.

SELECTED PROJECT

Hard Mass Conservation in an ML Aerosol Emulator | Python • PyTorch

github.com/SergioWatanabe/m7-mass-closure

- Reproduced the Harder et al. (2022) M7 aerosol microphysics neural network emulator, matching reported performance with test R^2 0.775 versus 0.763.
- Enforced exact mass conservation by orthogonal projection with residual about 1e-13 and no measurable accuracy loss across three seeds.
- Diagnosed a completion method that increased NMSE about 12 times on one channel, traced the cause to an approximately 2400 times cancellation ratio and identified data split and metric issues that understated violations by about five orders of magnitude.

TECHNICAL SKILLS

Programming: Python • **ML and scientific computing:** PyTorch, NumPy, pandas, scikit-learn, gradient-boosted trees, physics-constrained neural networks • **Tools:** Git, Docker • **Languages:** Portuguese, English (fluent)